

App Tutorial

A Simple Nextcloud App Tutorial



Nextcloud app tutorial
By Vitor Mattos



Quem sou eu?

Realizador de sonhos desde 2003

Amante de opensource

Palestrante

PHP Zend Certified Engineer ([ZEND024235](#))

PHPRio (<https://telegram.me/phprio>)

CTO LibreCode

Redes sociais: (VitorMattos ou VitorMattosRJ)



Command line tools

OCC



Command line tools

> alias

~/bashrc

```
if [ -f ~/.bash_aliases ]; then  
    . ~/.bash_aliases  
fi
```

~/bash_aliases

```
alias occ='docker compose exec -u www-data nextcloud ./occ'
```



Command line tools

> debug

```
occ config:system:get debug  
occ config:system:set debug 1
```



Command line tools

> xdebug

How to use xdebug on VSCode or Codium?

- Open the IDE on folder volumes/nextcloud
- setup the extension PHP Debug or PHP Extension Pack
- Press F5 to start debugging or go to "Run > Start debugging"
- Create a launch.json file to PHP
- Add the follow to your launch.json inside configuration named as Listen for Xdebug:

```
1 "pathMappings": {  
2   "src": "${workspaceFolder}": "${workspaceFolder}"
```

Types of apps

apps.nextcloud.com/



App architecture



- `appinfo/` Contains app metadata and configuration
- `css/` Contains the CSS
- `img/` Contains images and icons

Essentials

> appinfo/info.xml

```
1 <?xml version="1.0"?>
2 <info xmlns:xsi= "http://www.w3.org/2001/XMLSchema-instance"
3
4     xsi:noNamespaceSchemaLocation="https://apps.nextcloud.com/schema/apps/info.xsd">
5     <id>wifi_password</id>
6     <name>Wifi - Password</name>
7     <summary>Create, manage and share wi-fi passwords</summary>
8     <description>Create, manage and share wi-fi passwords using qrcode</description>
9     <version>1.0.0-alfa</version>
10    <licence>agpl</licence>
11    <author>Vinicius Reis</author>
12    <author>Vitor Mattos</author>
```



```
1 <?xml version="1.0"?>
2 <info xmlns:xsi= "http://www.w3.org/2001/XMLSchema-instance"
3
4     xsi:noNamespaceSchemaLocation="https://apps.nextcloud.com/schema/apps/info.xsd">
5     <id>wifi_password</id>
6     <name>Wifi - Password</name>
7     <summary>Create, manage and share wi-fi passwords</summary>
8     <description>Create, manage and share wi-fi passwords using qrcode</description>
9     <version>1.0.0-alfa</version>
10    <licence>agpl</licence>
11    <author>Vinicius Reis</author>
12    <author>Vitor Mattos</author>
13    <namespace>WifiPassword</namespace>
14    <category>files</category>
15    <category>tools</category>
16    <bugs>https://github.com/vinicius73/wifi-password/issues</bugs>
17    <dependencies>
18        <nextcloud min-version="22" max-version="25" />
```



Essentials

> lib/AppInfo/Application.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\AppInfo;
6
7 use OCP\AppFramework\App;
8
```



```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\AppInfo;
6
7 use OCP\AppFramework\App;
8
9 class Application extends App {
10     public const APP_ID = 'wifi_password';
11
12     public function __construct() {
13         parent::__construct(self::APP_ID);
14     }
15 }
```

Essentials

composer init

```
1 {
2     "name": "vinicius73/wifi-password",
3     "type": "project",
4     "license": "AGPL",
5     "authors": [
6         {
7             "name": "Vinicius Reis",
8             "email": "luiz.vinicius73@gmail.com"
9         },
10        {
```

```
1 {
2     "name": "vinicius73/wifi-password",
3     "type": "project",
4     "license": "AGPL",
5     "authors": [
6         {
7             "name": "Vinicius Reis",
8             "email": "luiz.vinicius73@gmail.com"
9         },
10        {
11            "name": "Vitor Mattos",
12            "email": "vitor@php.rio"
13        }
14    ],
15    "autoload": {
```



Essentials

composer init

Info: fix the namespace

```
1 {
2     "name": "vinicius73/wifi-password",
3     "type": "project",
4     "license": "AGPL",
5     "authors": [
6         {
7             "name": "Vinicius Reis",
8             "email": "luiz.vinicius73@gmail.com"
9         },
10        {
```



```
11         "name": "Vitor Mattos",
12         "email": "vitor@php.rio"
13     },
14 ],
15 "autoload": {
```

Enabling the app

```
occ app:enable wifi_password
```



Quality assurance

TDD



PRESENTING CODE

Quality assurance



PRESENTING CODE

Quality assurance composer-bin-plugin

```
composer require --dev bamarni/composer-bin-plugin
```

```
1 {  
2     ...  
3     "extra": {  
4         "bamarni-bin": {
```



```
5         "bin-links": true,  
6         "forward-command": true  
7     }  
8 }  
9 }
```

Quality assurance phpcs

```
composer bin coding-standard require --dev \
nextcloud/coding-standard
```

```
1 "scripts": {
```

```
2     "lint": "find . -name '*.php' -not -path './vendor/*' -
not -path './build/*' -print0 | xargs -0 -n1 php -l",
3     "cs:check": "php-cs-fixer fix --dry-run --diff",
4     "cs:fix": "php-cs-fixer fix"
5 }
```



Quality assurance phpcs

.php-cs-fixer.dist.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 require_once './vendor-bin/coding-standard/vendor/autoload.php';
6
7 use Nextcloud\CodingStandard\Config;
8
9 $config = new Config();
10 $config
11     ->getFinder()
```

```
12     ->ignoreVCSIgnored(true)
13     ->notPath('l10n')
14     ->notPath('src')
15     ->notPath('vendor')
```



Quality assurance psalm

```
composer bin psalm require --dev vimeo/psalm
composer bin psalm require --dev \
    nextcloud/ocp:dev-master
```

```
1 "scripts": {
2     "lint": "find . -name '*.php' -not -path './vendor/*' -
    not -path './build/*' -print0 | xargs -0 -n1 php -l",
```

```
3     "cs:check": "php-cs-fixer fix --dry-run --diff",
4     "cs:fix": "php-cs-fixer fix",
5     "psalm": "psalm",
6     "psalm:update-baseline": "psalm --threads=1 --update-
```



Quality assurance

psalm.xml

```
1 <?xml version="1.0"?>
2 <psalm
3     errorLevel="5"
4     resolveFromConfigFile="true"
5     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
6     xmlns="https://getpsalm.org/schema/config"
7     xsi:schemaLocation="https://getpsalm.org/schema/config
  vendor/vimeo/psalm/config.xsd"
8 >
9     <projectFiles>
10         <directory name="lib" />
11         <ignoreFiles>
12             <directory name="vendor" />
13         </ignoreFiles>
14     </projectFiles>
15     <issueHandlers>
16         <UndefinedClass>
17             <errorLevel type="suppress">
18                 <referencedClass name="OC\*" />
19                 <referencedClass name="OC" />
20             </errorLevel>
21         </UndefinedClass>
22     </issueHandlers>
23 </psalm>
24 <?xml version="1.0"?>
25 <psalm
26     errorLevel="5"
27     resolveFromConfigFile="true"
28     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
29     xmlns="https://getpsalm.org/schema/config"
30     xsi:schemaLocation="https://getpsalm.org/schema/config
  vendor/vimeo/psalm/config.xsd"
31 >
32     <projectFiles>
33         <directory name="lib" />
34         <ignoreFiles>
35             <directory name="vendor" />
36         </ignoreFiles>
37     </projectFiles>
38     <issueHandlers>
39         <UndefinedClass>
40             <errorLevel type="suppress">
41                 <referencedClass name="OC\*" />
42                 <referencedClass name="OC" />
43             </errorLevel>
44         </UndefinedClass>
45     </issueHandlers>
46 </psalm>
```



Quality assurance Behat

```
composer bin behat require --dev libresign/nextcloud-behat  
vendor/bin/behat init
```

behat.yml

```
default:  
  autoload:  
    '': '%paths.base%/tests/features/bootstrap'  
  suites:  
    default:
```

```
    paths:  
      - '%paths.base%/tests/features'  
  extensions:  
    PhpBuiltin\Server:
```



PRESENTING CODE

Quality assurance

Show me the code



Continuous Integration GitHub Actions

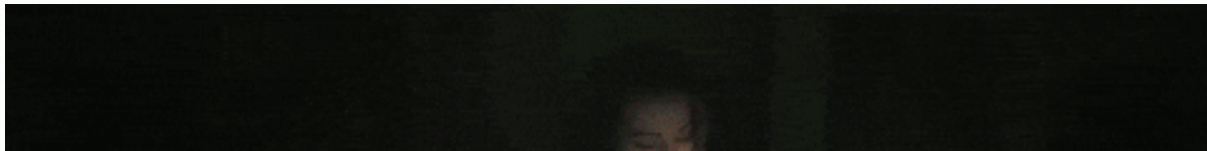
.github



PRESENTING CODE

Continuous Integration

Show me the code



First page

> appinfo/routes.php

```
1 <?php
2
3 return [
4     'routes' => [
5         ['name' => 'page#index', 'url' => '/', 'verb' => 'GET'],
6     ],
7 ];
```



Routes / API endpoints

connecting controller to route

Name

authorApi#someMethod



Routes / API endpoints

connecting controller to route

Split at #

and uppercase the first letter of the left part:

`authorApi`



Routes / API endpoints

connecting controller to route

Append Controller to the first part:

```
AuthorApiController
```



First page

> lib/Controller/PageController.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\Controller;
6
7 use OCP\AppFramework\Controller;
8 use OCP\AppFramework\Http\TemplateResponse;
9 use OCA\WifiPassword\AppInfo\Application;
10
11 class PageController extends Controller {
12     /**
```

```
13         * @NoAdminRequired
14         * @NoCSRFRequired
15         */
16     public function index(): TemplateResponse
17     {
```

First page
> templates/page-main.php

```
1 Hello World
```



First page Add to menu

```
1      ...
2      <dependencies>
3          <nextcloud min-version="26" max-version="26" />
4      </dependencies>
5      <navigations>
6          <navigation>
7              <name>Wifi Password</name>
```



```
8          <route>wifi_password.page.index</route>
9      </navigation>
10 </navigations>
11 </info>
```

First page
Add to menu

```
img/app.svg  
img/app-dark.svg
```



Routes / API endpoints

this...

```
1 return [  
2   'routes' => [  
3     ['name' => 'author#index', 'url' => '/authors', 'verb' => 'GET'],  
4     ['name' => 'author#show', 'url' => '/authors/{id}', 'verb' => 'GET'],  
5     ['name' => 'author#create', 'url' => '/authors', 'verb' => 'POST'],  
6     ['name' => 'author#update', 'url' => '/authors/{id}', 'verb' => 'PUT'],  
7     ['name' => 'author#destroy', 'url' => '/authors/{id}', 'verb' => 'DELETE'],  
8     // your other routes here  
9   ],  
10  ];
```



API endpoints

can be abbreviated by using the **resources** key:

```
1 return [  
2   'resources' => [  
3     'author' => ['url' => '/authors'],  
4   ],  
5   'routes' => [  
6     // your other routes here
```



```
7   ],  
8 ];
```

PRESENTING CODE

API
Tip



Insomnia



Getting request parameters

Parameters can be passed in many ways:

- Extracted from the URL using curly braces: `{key}` inside the URL: `Routing`
- Appended to the URL: `key=value` or `key=value&key=value`



Getting request parameters

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
9     /**
10      * @param int $id
11      */
```

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
9     /**
10      * @param int $id
11      */
12     public function doSomething(int $id, string $name='john', string $job='author'):
13         Response {
14         // GET ?id=3&job=Developer
15         // $id = 3
16         // $name = 'john'
17         // $job = 'Developer'
```



Getting request parameters

json

```
1 POST /index.php/apps/myapp/authors
2 Content-Type: application/json
3 {
4     "name": "test",
5     "number": 3,
6     "publisher": true,
7     "customFields": {
```

```
1 POST /index.php/apps/myapp/authors
2 Content-Type: application/json
3 {
4     "name": "test",
5     "number": 3,
6     "publisher": true,
7     "customFields": {
8         "mail": "test@example.com",
9         "address": "Somewhere"
10    }
11 }
```



Getting request parameters json

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
```

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
9     public function create(string $name, int $number, string
    $publisher, array $customFields): Response {
10         // $name = 'test'
11         // $number = 3
12         // $publisher = true
13         // $customFields = array("mail" => "test@example.com",
    "address" => "Somewhere")
```



Getting anything else

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6 use OCP\IRequest;
7
8 class PageController extends Controller {
9
10     public function someMethod(): Response {
11         $type = $this->request->getHeader('Content-Type'); //
12         $_SERVER['HTTP_CONTENT_TYPE']
13         $cookie = $this->request->getCookie('myCookie'); //
14         $_COOKIES['myCookie']
15     }
16 }
17 }
```



```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6 use OCP\IRequest;
7
8 class PageController extends Controller {
9
10     public function someMethod(): Response {
11         $type = $this->request->getHeader('Content-Type'); //
12         $_SERVER['HTTP_CONTENT_TYPE']
13         $cookie = $this->request->getCookie('myCookie'); //
14         $_COOKIES['myCookie']
15         $file = $this->request->getUploadedFile('myfile'); // $_FILES['myfile']
16         $env = $this->request->getEnv('SOME_VAR'); // $_ENV['SOME_VAR']
17     }
18 }
```



More about controllers

Session
Cookies
and more...

https://docs.nextcloud.com/server/latest/developer_manual/basics/controllers.html



PRESENTING CODE

Dependency injection



symfony/**dependency-**



Dependency injection

Don't put new dependencies in your constructor or methods but pass them in.

So this...

```
1 <?php
2 use OCP\IDBConnection;
3
4 // without dependency injection
```

```
5 class AuthorMapper {
6     private IDBConnection $db;
7
8     public function __construct() {
9         $this->db = new Db();
10    }
```



Dependency injection

would turn into this by using Dependency Injection:

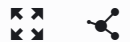
```
1 <?php
2
3 use OCP\IDBConnection;
4
```



```
5 // with dependency injection
6 class AuthorMapper {
7     private IDBConnection $db;
8
9     public function __construct(IDBConnection $db) {
10         $this->db = $db;
11     }
12 }
```

Middlewares

Middleware is logic that is run before and after each request



https://docs.nextcloud.com/server/latest/developer_manual/basics/middlewares.html

Events

Communicate between different aspects of the Nextcloud eco system.

Implement a Listener to intercept events



Migrations

> occ commands



- `migrations:execute`: Executes a single migration version manually.
- `migrations:generate`: Create a new migration file.
- `migrations:migrate`: Execute all pending migrations.

Migrations

> creating new migration



```
occ migrations:migrate <appid> <version>
```



Migrations

> creating new migration

Migration was named:

Version<version>Date<date>



Example

www.dbooks.org

Migrations

> creating new migration

Version

I recommend to use Semantic Version without dots.



Example

v1.2.0

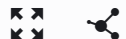
1.2.0

Frontend

> Adding scripts

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\Controller;
6
7 use OCP\AppFramework\Controller;
8 use OCP\AppFramework\Http\TemplateResponse;
9 use OCA\WifiPassword\AppInfo\Application;
10
11 class PageController extends Controller {
12     /**
```

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\Controller;
6
7 use OCP\AppFramework\Controller;
8 use OCP\AppFramework\Http\TemplateResponse;
9 use OCA\WifiPassword\AppInfo\Application;
10
11 class PageController extends Controller {
12     /**
13      * @NoAdminRequired
14      * @NoCSRFRequired
15      */
16     public function index(): TemplateResponse
17     {
18         Util::addScript(Application::APP_ID, 'page-main');
```



Frontend

> Initial state

```
1 class PageController extends Controller {
2     private IInitialState $initialState;
3     public function __construct(
4         IRequest $request,
5         IInitialState $initialState
6     ) {
7         parent::__construct(Application::APP_ID, $request);
8         $this->initialState = $initialState;
9     }
10
11     /**
12      * @NoAdminRequired
```

```
1 class PageController extends Controller {
2     private IInitialState $initialState;
3     public function __construct(
4         IRequest $request,
5         IInitialState $initialState
6     ) {
7         parent::__construct(Application::APP_ID, $request);
8         $this->initialState = $initialState;
9     }
10
11     /**
12      * @NoAdminRequired
13      * @NoCSRFRequired
14      */
15     public function index(): TemplateResponse
16     {
17         Util::addScript(Application::APP_ID, 'page-main');
18         $this->initialState->provideInitialState('config', 'value');
```



Frontend

> Initial state

Press F12 and check the input with all initial states

Load in frontend:



https://docs.nextcloud.com/server/latest/developer_manual/basics/frontend/js.html#loading-initial-state

PRESENTING CODE

Frontend

> Vue Components



Analyzing the code

<https://github.com/nextcloud/app-tutorial>

[https://docs.nextcloud.com/server/latest/developer_manual/app_development/tutorial.ht](https://docs.nextcloud.com/server/latest/developer_manual/app_development/tutorial.html)



Obrigado!

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